

Education

University of California, Santa Cruz

Doctor of Philosophy

Santa Cruz, CA
Sept. 2022 – Present

Advisor: Dr. Nia Imara

- Investigating Giant Molecular Cloud (GMC) formation in the Milky Way
- Segmenting galaxy-wide survey datasets to create molecular cloud and atomic cloud catalogs

Pomona College

Bachelor of Arts

Claremont, CA
May 2021

Major: Astrophysics & Philosophy

Minor: Studio Art

- GPA: 3.8
- Self-designed a major that aims to form a deep understanding of how the universe functions, and simultaneously examine current scientific paradigms through the lenses of epistemology and philosophy of science

Research and Work Experience

University of California, Santa Cruz, Teaching Assistant

Santa Cruz, CA
April – June 2023

- Assisted with an upper division undergraduate course on the Physics of Stars
- Provided support during in-class exercises
- Conducted sections to supplement lecture material and provide homework help
- Graded homework, proctored midterms and exams

Pomona College, Post-Baccalaureate Lab Coordinator

Claremont, CA
Aug. 2021 – June 2022

- Hosted tutorial-style pre-lab sessions and taught students the background theory for experiments in an introductory mechanics course
- Re-calibrated echelle spectrograph and gathered preliminary data
- Organized meetings for racial and gender minorities within Physics and Astronomy

Pomona College, Gravitational Wave Simulation Research Assistant

Claremont, CA
June – Aug. 2021

- Contributed to model that simulates the detections of gravitational waves emitted by a binary star system
- Implemented various numerical methods (Runge Kutta, Leap Frog) into the simulation
- Compared results between different algorithms and alter the program to increase efficiency
- Wrote a training manual for future research assistants, including documentation for the program and a summary of 10 years' worth of past work on the project

The Women+ of Color Project, Administrative Assistant

Remote/Boston, MA
April 2021 – Sept. 2022

- Organized a graduate school application workshop for racial/gender minorities in STEM
- Arranged meetings between committees, as well as with external contacts
- Responded to inquiry emails with the relevant information

Pomona College, Academic Mentor

Claremont, CA
Aug. 2018 – May 2021

- Aided students in understanding physics concepts, formulas, and problem solving strategies
- Supported students during remote observing labs
- Courses: Stellar Mechanics, Planetary Motion and Basic Observational Skills, Mechanics, Special Relativity, Quantum Mechanics, Electricity and Magnetism, Thermodynamics, and Philosophy of Physics

Office of Astronomy for Development (OAD), Summer Intern

Remote/Cape Town,
South Africa
July 2020

- Collected and cleaned data to be used in a hackathon for high schoolers
- Sorted data according to affective qualities for use in training a machine learning algorithm
- Developed a year-long research project exploring the intersections of astronomy and mental health

Jet Propulsion Laboratories (JPL), Remote Observer

Pasadena, CA
Aug 2019 – Mar 2020

- Observed and confirmed the trajectories of 50+ asteroids and near earth objects
- Created finding charts and ephemerides for asteroids using JPL Horizons and Minor Planet Catalog
- Troubleshoot hardware malfunctions such as dome-telescope decoupling
- Worked with a team of other researchers to coordinate schedules, share observational data

Conference Presentations

“Developing an Algorithm to Model the Evolution of a Binary Star System,” *National Society of Black Physicists (NSBP)*; Virtual

November 2021

“Developing an Algorithm to Model the Evolution of a Binary Star System,” *Conference for Undergraduate Women in Physics*; University of Nebraska, Lincoln

October 2021

Awards Received

- Pomona College Scholar (2017, 2021)
- Distinction in Senior Exercise (2021)